CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. ₹ 1900 is divided in parts proportional to $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{5}$. The largest part is: 1

(a) ₹ 1000 **(b)** ₹ 500 **(c)** ₹ 400 **(d)** ₹ 380

2. 10 men can finish a job in 24 days. After 6 days, 5 more men join. The remaining work will be finished in: 1

(a) 18 days **(b)** 14 days **(c)** 12 days **(d)** 16 days

3. Two trains cover the same distance at speeds in the ratio 3 : 7. The ratio of the times they take is: 1

(a) 3 : 7 **(b)** 9 : 49 **(c)** 7 : 3 **(d)** 10 : 3

4. If 2 : 7 = 6 : 21, then (2+7) : 7 = x : 21, where x is: 1

(a) 21 **(b)** 27 **(c)** 34 **(d)** 6

5. The sides of two squares are in the ratio 3 : 5. The ratio of their areas is: 1

(a) 3 : 25 **(b)** 9 : 25 **(c)** 3 : 5 **(d)** 9 : 5

6. The number x for which 6 : 5 = 18 : x is: 1

(a) 18 **(b)** 15 **(c)** 23 **(d)** 29

7. If x : y = 5 : 6 and y : z = 9 : 4, then x : y : z is: 1

(a) 5 : 6 : 4 **(b)** 45 : 54 : 24 **(c)** 5 : 15 : 4 **(d)** 15 : 18 : 8

8. The present ages of two brothers are in the ratio 4 : 3. After 5 years the ratio becomes 5 : 4. The present age of the elder brother is: 1

(a) 15 years **(b)** 25 years **(c)** 35 years **(d)** 20 years

9. **Assertion (A):** If 6 pens cost ₹90, then 10 pens cost ₹150. 1
Reason (R): In the unitary method we first find the cost of 10 pens.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The ratio $2 : 3$ is greater than the ratio $3 : 4$. 1
Reason (R): Ratios can be compared by converting them to fractions with the same denominator.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If x : y = 9 : 5, find the ratio (x + y) : (x - y). 2

12. Divide ₹ 1400 into parts proportional to $\frac{1}{3}$ and $\frac{1}{4}$. 2

13. The sides of two squares are in the ratio 6 : 11. Find the ratio of their perimeters and of their areas. 2

14. If x : y = 11 : 4, find the ratio (x + y) : (x - y). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The ratio of the present ages of a father and son is 4 : 1. After 10 years the ratio will be 5 : 2. Find their present ages. 3

16. 16 men can finish a job in 18 days. After 6 days, 8 more men join them. In how many more days is the job finished? 3

17. Divide ₹ 2100 into three parts proportional to $\frac{1}{3}$, $\frac{1}{5}$ and $\frac{1}{6}$. 3

OR

The ratio of the present ages of a father and son is 7 : 3. After 2 years the ratio will be 15 : 7. Find their present ages. 3

18. 18 men can finish a job in 20 days. After 4 days, 6 more men join them. In how many more days is the job finished? 3

OR

Divide ₹ 3760 into three parts proportional to $\frac{1}{2}$, $\frac{1}{5}$ and $\frac{1}{10}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A mason in Kota earns ₹4,500 for 6 days of work, while another mason earns ₹3,900 for 5 days. Assume daily wages are fixed. 5
(i) Find the first mason's wage per day.
(ii) How much does he earn in 10 days?
(iii) In how many days does he earn ₹12,000?
(iv) Who earns more per day, and by how much?

20. A, B and C invest ₹ 4000, ₹ 5000 and ₹ 3000 for 6, 4 and 8 months respectively. A profit of ₹ 1360 is shared in the ratio of (investment × time). Find the ratio and each share. 5

OR

A vessel holds 100 L of a mixture of milk and water in the ratio 7 : 3. 10 L of the mixture is removed and replaced by 10 L of water. Find the new ratio of milk to water. 5

END OF QUESTION PAPER — **Total: 40 Marks**